

MANGALORE REFINERY & PETROCHEMICALS LIMITED

Standard Operating Procedure: K-101 Centrifugal
Recycle Gas Compressor

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Target Equipment: K-101 (Recycle Gas Compressor)	Suction Valve: MOV-101 (Isolation Actuator)
Operating Unit: Unit 01 - Hydrotreater	Process Medium: High-Pressure Hydrogen / Light Hydrocarbon Gas
Author / Engineer: Vicky S. (Lead Reliability)	Approving Supervisor: Jane D. / Vicky S. (Four-Eyes)

1. Equipment Description & Operational Envelope

Compressor **K-101** is a multi-stage centrifugal recycle gas compressor driven by a 4,200 kW, 6.6 kV electric motor (M-101). The machine recirculates hydrogen-rich treat gas through the Unit-01 hydrotreating reactors. All operators and automated control loops must adhere strictly to the safe operating envelope established below:

Parameter	Normal Operating Envelope	Warning Alarm	Trip Limit (Emergency Trip)
Suction Pressure (PT-100)	30.0 - 35.0 bar(g)	< 28.0 or > 37.0 bar(g)	< 25.0 or > 40.0 bar(g)
Discharge Pressure (PT-101)	64.0 - 70.0 bar(g)	> 72.5 bar(g)	> 76.0 bar(g) (Relief Valve 78.0)
Operating Speed (ST-101)	10,800 - 11,400 RPM	< 10,200 or > 11,800 RPM	> 12,200 RPM (Overspeed Trip)
Radial Vibration (VT-102A/B)	1.2 - 2.8 mm/s RMS	> 3.5 mm/s RMS	> 4.5 mm/s RMS
Bearing Temp DE (TT-204)	72.0 - 88.0 °C	> 90.0 °C	> 95.0 °C
Lube Oil Supply Press (PT-108)	4.6 - 5.4 bar(g)	< 4.0 bar(g)	< 3.2 bar(g)
Dry Gas Seal DP (PDIT-101)	3.8 - 4.8 bar delta	< 3.2 bar delta	< 2.8 bar delta

2. Pre-Startup Verification & Permitting Protocol

Prior to issuing any start command to compressor K-101, the shift lead technician and DCS console operator must complete the following mandatory checks:

- **Lube Oil System:** Ensure auxiliary lube oil pump is running and lube oil temperature is between 38°C and 45°C.
- **Seal Gas Supply:** Confirm nitrogen barrier gas and primary seal gas differential pressure (PDIT-101) exceeds 3.5 bar delta across all seals.
- **Valve Lineup:** Confirm motor-operated suction isolation valve **MOV-101** is in remote-auto mode, fully stroked, and confirmed OPEN.
- **Cooling Water:** Verify shell-and-tube heat exchanger cooling water flow is verified at nominal 120 m³/h.
- **Interlock Clearance:** Confirm all 14 safety trip channels report HEALTHY on the ESD-101 Safety Instrumented System.

3. Four-Eyes Authorization & High-Risk Interventions

In compliance with SIH-26117 air-gapped industrial governance, any modification of operating parameters, service restart, or emergency bypass of compressor K-101 requires strict **Human-in-the-Loop (HITL) Dual Supervisor Authorization (Four-Eyes Principle)**.

MANDATORY DUAL AUTHORIZATION POLICY:

1. Tool restart_service for K-101 or M-101 is classified as service_interrupting.
2. Automated single-agent or single-operator execution is strictly prohibited by system interlocks.
3. The action must be submitted via the CogniShift Authorizations API and digitally signed by two independent credentialed supervisors (e.g. Vicky S. and Jane D.) before the DCS permit is released.

4. Emergency Trip & Flare Venting Protocol

In the event of uncontrolled vibration (> 4.5 mm/s RMS) or loss of seal gas differential (< 2.8 bar delta), the automated trip interlock or manual operator emergency trip button will trigger the following fail-closed sequence:

1. **Motor Trip:** Breaker 52-K101 opens in under 40 milliseconds, cutting 6.6 kV feed to drive motor M-101.
2. **Suction Isolation:** Actuator **MOV-101** closes immediately with a stroke time under 6.0 seconds to prevent reverse flow.
3. **Emergency Venting:** Fast-acting depressurization valve XV-101 actuates to vent the compressor casing gas to the **Flare-Header** (High-Pressure Acid Gas Flare Header).
4. **Nitrogen Purge:** Emergency nitrogen sweep valve XV-105 opens automatically to dilute hydrocarbon concentrations below Lower Explosive Limit (LEL).

5. Diagnostics & Troubleshooting Matrix

Observed Symptom	Probable Root Cause	Automated Sensor Tag	Prescribed Corrective Action
Radial vibration excursion > 3.8 mm/s on DE	Liquid carryover into 1st stage impeller or coupling misalignment	VT-102A / VT-102B	Check suction KO drum level (LT-104); drain condensate immediately.
Bearing temp DE rising above 90.0 °C	Lube oil cooler fouled or low oil supply flow rate	TT-204 / PT-108	Switch dual lube oil filter basket; increase cooling water bypass.
Seal gas DP dropping below 3.2 bar delta	Primary vent orifice restriction or nitrogen booster failure	PDIT-101 / PI-105	Verify nitrogen backup supply cylinder manifold pressure.
MOV-101 fails to reach fully closed state	Actuator torque limit switch trip or stem mechanical obstruction	ZSO-101 / ZSC-101	Engage manual handwheel declutch lever; verify mechanical stem travel.